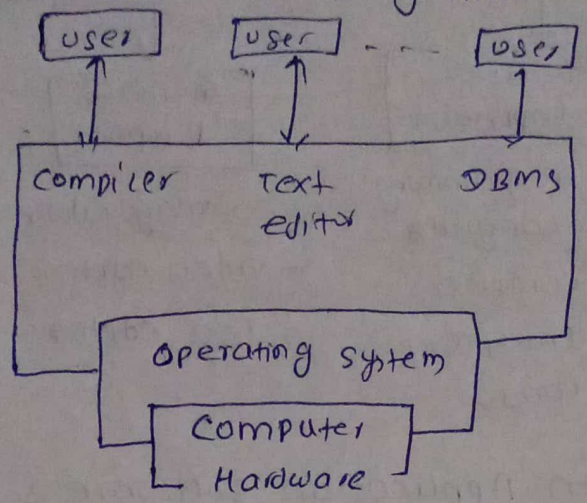
OS

UNIT - I

Operating system is a program that acts as an interface between user and computer hardware and it controls execution of all kinds of programs



A computer system can be roughly divided into 4 components Hardwae, operating system, Application software and users.

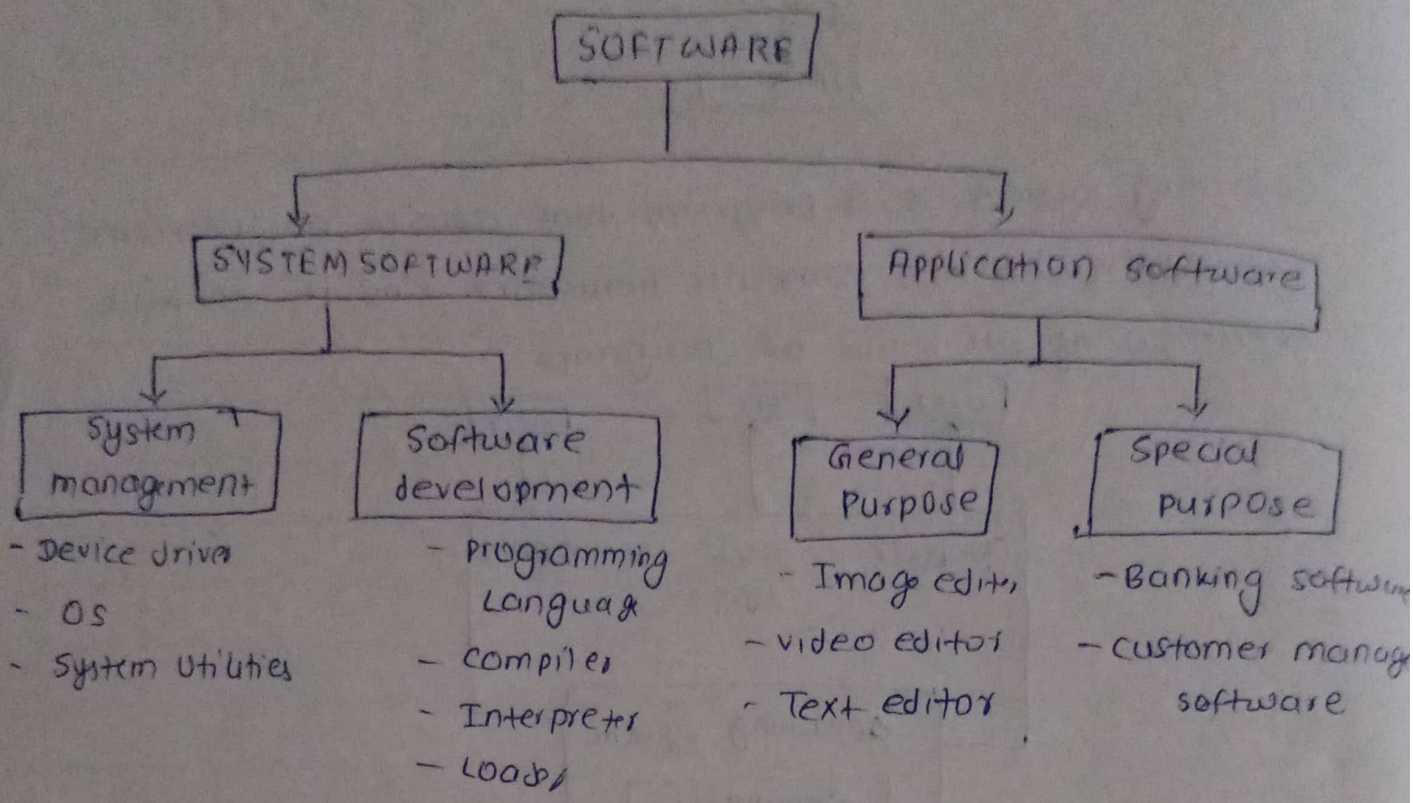
- Hardware includes central proce. unit (CPU), memory, I/o devices etc.
- Application programs define ways in which resources are used to solve computing problems of the users.
- Operating system controles and cordinates the use of Hardware among various application programs.

Examples of OS:-

LINUX, UNIX, MAC, WINDOWS XP, 7, 8, 10, 11

IBM AIX

VMS OS/400



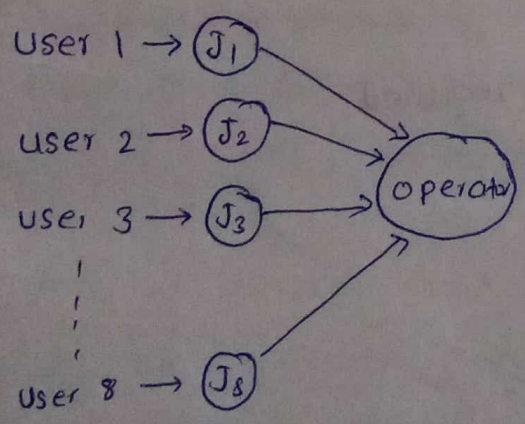
* Difference between Application software & system software :

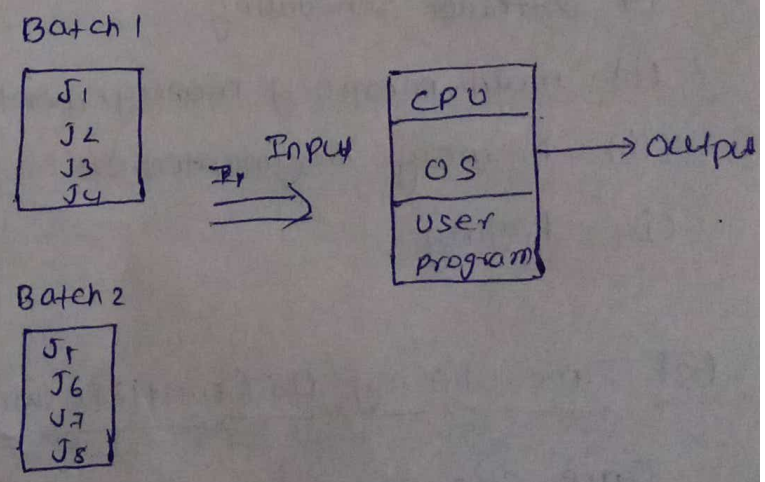
System software	Application software
- It is used for the management of computer.	- It is used to perform specific tasks.
- These are installed on the computer when OS is installed.	- These are installed according to user requirements.
- User does not interact with system software becaus it works in background.	- User interacts with application software.
- System software can run independently. It provides platform for running application software.	- It can not run independently. It can not run without the presence of system software.

* Classification of Operating system:-

(1) Batch operating system:-

- Users of Batch operating system do not interact with the computer directly. Each user prepares his job on an offline device like punch card and submit it to the computer operator.
- To speed up processing similar tasks are grouped together as a batch.
- Programmers leave their programs with the operator and then operator executes similar programms together in a batch.
- The problem with batch operating system are -
  (i) Lack of interaction b/w the user and the system.
  (ii) Less CPU utilisation because speed of mechanic I/O devices is very slow in comparison of CPU.





Job = Input data + control instructions.

(2) Multiprogrammed OS :-

- Multiprogramming is a technique to store more than one process in main memory which are ready to execute.
- Process generally requires CPU time and I/O time so if running process performs I/O or some other event which do not require CPU then instead of seating idle CPU makes a context switch and select some other process for execution. So CPU utilisation will improve

Advantages :-

(i) High CPU utilisation

(ii) Less waiting/response time

(iii) This concept can be extended to multiple users.

Disadvantages :-

(i) ~~Different~~ scheduling

(ii) Main memory management is required

(iii) Memory fragmentation

(iv) Paging

(3) Time sharing OS (multitasking) :-

Time sharing or multitasking is logical extension of multiprogramming. Processors time which is shared among multiple users simultaneously is termed as time sharing.

- The main difference between multiprogrammed OS and time sharing OS is that in case of multiprogrammed OS, objective is to maximise processor use whereas in case of time sharing system the objective is to minimise response time.
- In time sharing OS task are executed by the CPU by switching between them but the switches happens very frequently.
- The time sharing OS uses CPU scheduling and multiprogramming to provide each user small amount of time so multitasking = multiprogramming + context switching.
- Time sharing OS is also referred as "multiprogrammed with Round Robin".

Advantages of Time sharing OS :-

(i) Reduces CPU Idle time

(ii) Provides Improved Response time

(4). Real time OS :- (Interactive) :-

- Real time OS are used in environment where a large no. of events must be accepted and processed in short-time or within certain dead lines. So, this system is time bound and has a fixed dead line.
- The time taken by the system to respond to an Input and display required updated information is termed as Response time. Further real time OS is classified into two parts: (i) Hard Real time OS

(ii) Soft Real time OS

(i) Hard Real time OS :-

These OS are used in time bound critical applications.

For ex:- Airline reservation, Air traffic control system, medical equipments (ex- Heart disease equipments), different applications, share market etc.

(ii) Soft Real time OS :-

- These are less restrictive.
- These systems have limited utility as compared with Hard real time OS.

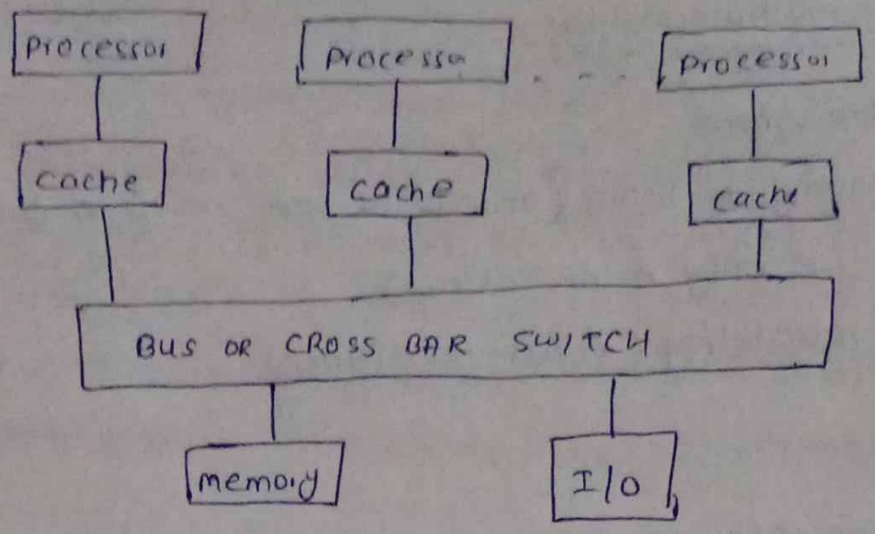
For ex:- Gaming application

* MULTIPROCESSOR SYSTEM :-

- It refers to use of two or more CPU within single computer system. These multiple CPU are in close communication and shares computer bus memory and other peripheral devices. These systems are reffered as Tightly-Coupled multiprocessor OS.
- In multiprocessing multiple concurrent process run on a seperate CPU. These types of systems achieve true parallel execution of processes.

Applications :- weather forecasting
Image processing
AI System
Expert system

Expert system uses AI techniques to similitate judgement and behaviour of a human.



* Coupling:- Degree of interdependance among software modules.

* Tightly coupled multiprocessor OS:-

Degree of coupling b/w processors is High.

* Loosely coupled multiprocessor OS:-

Degree of coupling b/w processor is Low.

* Differentiate b/w Loosely & tightly coupled OS:-

Parameters	Loosely coupled	tightly coupled
(i) Memory	Each processor has its own memory module.	(i) processors have shared memory modules.
(ii) memory conflict	It generally do not encounter memory conflict.	It experiences frequent memory conflicts.
(iii) Interconnection network	Message Transfer System (MTS)	Interconn. network such as PMIN, IO PIN, ISIN.
(iv) cost	Less Expensive	More expensive
(v) Efficiency	Efficient when task running on diff. processors have very less interaction.	Efficient for high speed or real time processor.

* Advantages of Multiprocessing:-

(i) Increased Throughput

(ii) Provides economy of scale (instead of replicating n different systems we use only n processors).

(iii) Increased reliability (Fault-tolerance)

Disadvantages:-

(i) complexity increases as compared with uni processor OS

(ii) Requires complicated context switching.

Multiprocessor	Multi-core
- A system with two or more CPU that allows simultaneous execution of processes.	- Single CPU with two or more processing elements that are capable of reading and executing program instructions.

** Functions of OPERATING SYSTEM.

(1) Memory management-

- It refers to management of primary memory or main memory.
- Main memory is a large array of bytes or words where each word has a specific memory address.
- Main memory provides a fast storage that can be accessed directly by the CPU.
- A program must reside in main memory in order to be executed.
- An OS does following activities for memory management -

(i) keeps track of primary memory.

(ii) In multiprogramming the OS decides which process will get memory when and how much.

(iii) When a process request for execution OS allocates memory to it.

(iv) Re-allocates the memory when a process has been terminated.

(2). Process management:-

In multiprogramming environment operating system decides which process gets the CPU and for how much time. This function is called process scheduling. An operating system performs following activities for process management:-

(i) Keeps track of processor and status of the process. The program responsible for this task is called traffic controller.

(2). Allocates CPU to the process.

(3). Deallocates processor when process has been terminated.

(3). Device management:-

OS performs following activities for device management:-

(i) Keeps track of all the devices. The program responsible for this task is called as I/O controller.

(ii) Besides Decides which process gets the device and for how much time.

(iii) Allocates and deallocates the device.

(4). File management:-

An OS performs following activities for file management:-

(i) keeps track of information such as location, users, status etc.

(ii) Allocates and deallocates the resources.

(5). Other important activities of OS:-

(i) Managemt of software and users.

(ii) Security:-

OS prevents unauthorised access to program and data by using passwords and user authentication techniques.

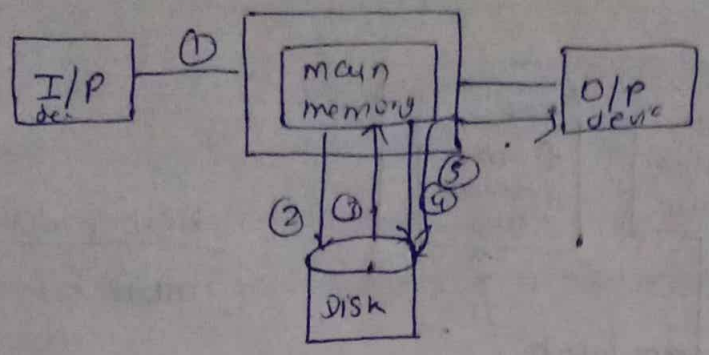
(iii) Error detection and correction.

OS provides error messages, debugging and error detecting techniques in order to handle with errors and interrupts.

* SPOOLING:-

- It stands for simultaneous peripheral operations online.
- Spooling is a process in which data of various input/output operations is stored in a special area of memory or hard disk which is accessible to I/O device.
- Spooling overlaps operation of one task with execution of another task.
- I/O devices are comparatively slow in comparison of CPU so when CPU is processing that time I/O operations is performed for another task to improve system performance.

Ex:- Print spooler is a software build into windows OS that stores print commands temporarily in computers memory until printer is ready to print them.



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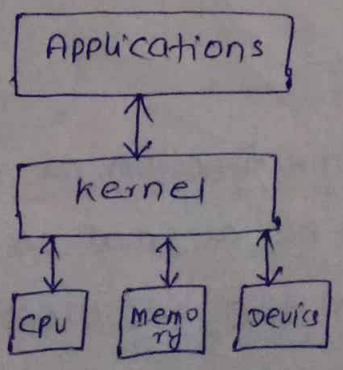
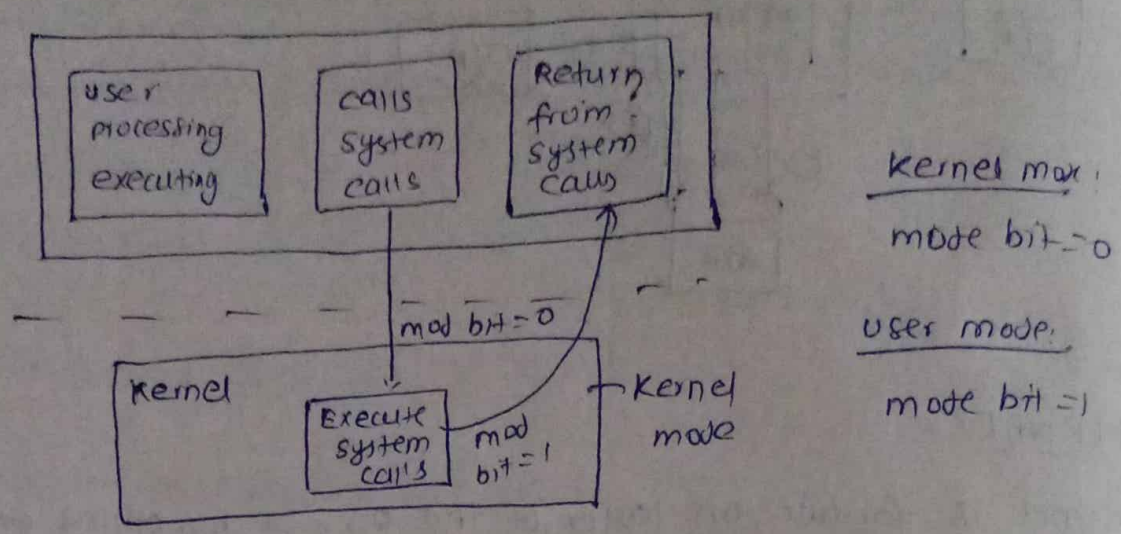
KERNEL :-

- kernel is foundation layer of the OS. It functions ~~and~~ at basic level communicating with hardware and managing resources such as RAM.
- Since a kernel handles many fundamental processes it must be loaded at the beginning when a computer starts.
- The kernel performs system check and recognises components such as processor and memory. It also checks for any connected peripherals.
- As the operating system loads and practical user interface appears the kernel keeps running.
- Even after operating system has fully loaded the kernel continues to run in the background. managing system resources.



1. General description of kernel



Types of KERNEL :-

Several types of kernels exists but two popular kinds are

(1) Monolithic kernel

(2) Micro kernel

Another category of kernel is Re-entrant kernel.

(1) Monolithic kernel :-

- It is the oldest architecture used for developing OS.
- It envolves simple design and creates well-defined communi, layer between hardware and software.
- A monolithic kernel is a single block of source code that provides all necessary services provided by operating system.

(2) Micro kernel :-

- Microkernels have the same function as monolithic kernel but they are designed to be as small as possible.
- Instead of managing all the resources from single code file the microkernel handles only the most basic function. (scheduling and Inter process communication).
- It uses modules to manage other functionality of OS.
- The basic idea behind microkernels is to remove all non-essential components from the kernel and implement them as system applications, thereby making kernel as small

and efficient as possible.

- Most microkernel's provide basic process and memory management and message passing between other services.
- Security and the protection can be enhanced as most services are performed in user mode and not in kernel mode.
- System expansion becomes easier as it envolves adding more system applications and not rebuilding the new kernel.
- For ex:- device drivers are typically included in monolithic kernel but they would be split into seperate modules in microkernels.
- This design is more complex but it can provide more efficient use of system resources and helps against system crashes.

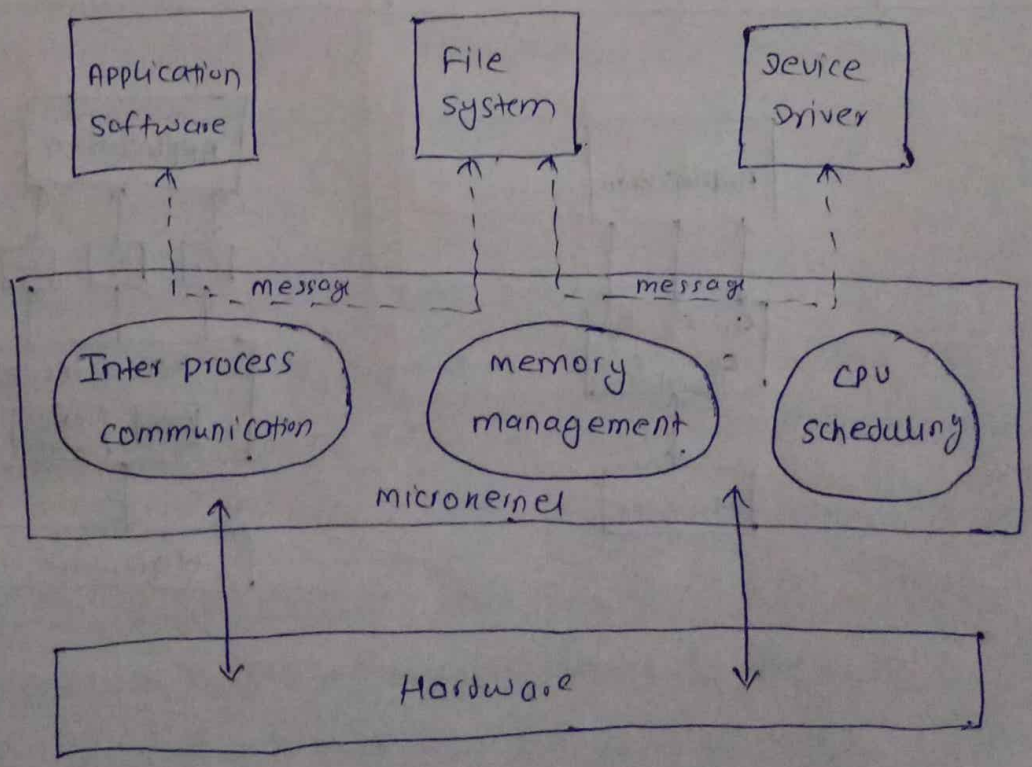
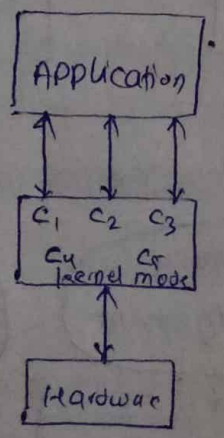


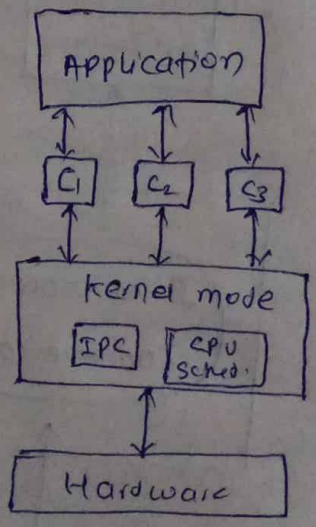
Fig: Architecture of microkernel.

* Difference b/w monolithic and microkernel :-

~~Monolithic kern~~

Basis for comparison	Monolitic	microkernel
1. Size	It is larger than micro kernel.	It is smaller than monolithic.
2. Execution	Fast execution	Slow execution because comm. b/w system applications is established using message passing.
3. Security	If a service crashes then entire system will crash.	If a service crashes it does not effect the working of micro kernel.
	Ex: MS-DOS Solaris	windows OS MAC OS





* Operating System services:-

Operating system provides services to both users and to the programs. Following are few common services provided by OS:-

(i) Program Execution
(ii) I/o operation
(iii) File system manipulation.
(iv) Communication.
(v) Error detection
(vi) Resource allocation.
(vii) Protection and security.

(1) Program Execution:-

(i) Loads a program into memory
(ii) Executes the program
(iii) Handles program's execution.
(iv) Provides a mechanism for process synchronization.
(v) Provides mechanism for deadlock handling.

(2) Error handling:-

Error may take place in CPU, I/o device or in memory hardware. Operating system constantly checks the possible errors. Operating system takes appropriate action to ensure correct functioning.

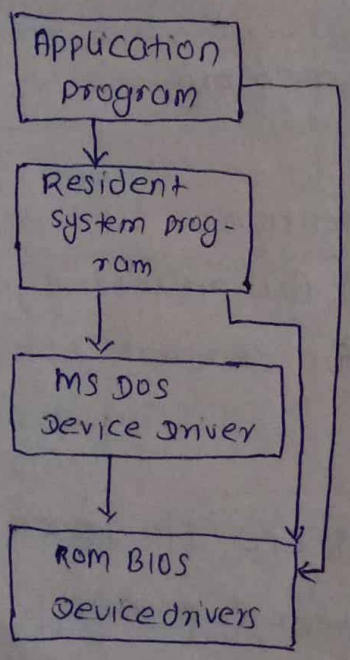
(3) Resource management:-

OS manages all kinds of resources using scheduler. CPU scheduling Algorithms like FCFS, SJF, Round Robin are used for better utilisation of CPU.

* OS Structure / OS Architecture :-

(i) Simple structure :-

- MS DOS has very simple structure and most functionality is written in single least space.
- In this case levels of the functionality are not well seperated.
- There is no CPU execution mode, so error in application could lead to the entire system crash.
- MS DOS allows the users to navigate and manipulate files using command line.



(ii) Layered Approach :-

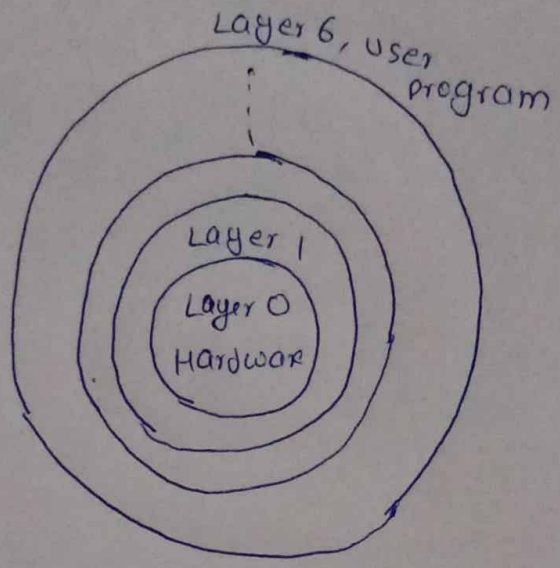
- In layered approach, OS is split into a number of layers where each layer is developed and debug independently. Each layer relise on the seruices of the layer below it.
- Communication takes place only between consecutive layers.
- Example of layered OS with 6 layers can be given as -

Layer 1 : Hardware

Layer 2 : CPU scheduling

Layer 3 : Memory management.

Layer 4 : Process management.

Layer 5 : I/O Buffer

Layer 6 : User programs.